



GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. IV Semester, I-Midterm Exam 2025-26

4CS2-01/4IT2-01/4AID2-01/4CYB-01/4IOT2-01: Discrete Mathematics Structure
CSE/IT/AI&DS/Cyber/IOT
09/03/2026/MONDAY

Time: 2 Hours

Maximum Marks: 40

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Part A (Answer should be given up to 25 words only)

All questions are compulsory

10x1 = 10

- Q.1 Define the Relation. [CO1,L1]
 Q.2 What is the domain of the function $f(x) = \frac{x}{x^2+3x+2}$ [CO1,L1]
 Q.3 Write the power set of the set $A = \{x, y, z\}$ [CO1,L3]
 Q.4 Define the symmetric difference of two sets using Venn diagram? [CO2,L1]
 Q.5 Write the properties of equivalence relation. [CO1,L3]
 Q.6 If $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{2, 4, 6, 8\}$ then Find $B - A$. [CO1,L1]
 Q.7 State the Pigeonhole principal. [CO2,L1]
 Q.8 Define onto function with example. [CO1,L1]
 Q.9 Define inverse function with example. [CO1,L1]
 Q.10 If $f(x) = e^x$ and $g(x) = x^3$ then find the composition function $f \circ g$. [CO2,L1]

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

5 x 3 = 15

- Q.1 Prove that $\sqrt{2}$ is irrational number. [CO2,L2]
 Q.2 Draw the Venn Diagrams of the following $(A \cup B)$, $(A \cap B)$ and Complement of A. [CO1,L1]
 Q.3 Prove by principal of mathematical induction For all $n \geq 1$, prove that, $1^3 + 2^3 + 3^3 + \dots + n^3 = \{n(n+1)/2\}^2$ [CO2,L2]
 Q.4 Prove that: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ [CO2,L2]
 Q.5 Let Q is the rational number. Show that the function $f: Q \rightarrow Q$ defined by $f(x) = 3x+7$, $x \in Q$ is a Bijective function. [CO1,L3]

Part C (Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions (word Limit 125)

3x 5 = 15

- Q.1 Let 100 out of 120 students of mathematics at a college take at least one of the language Hindi, English and German. Also let 65 study Hindi, 45 study English and 42 German. If 20 study Hindi and English, 25 study English and German and 15 study Hindi and German. Find the number of students whose study all the three languages. [CO2,L1]
 Q.2. Prove that $5^n + 3$ is divisible by 4 for all integers $n \geq 0$. [CO2,L2]
 Q.3. Define the Following (a) Subset, (b) Universal set, (c) one to one relation, (d) Countable set (e) Inverse Relation [CO1,L1]



GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. IV Semester, II-Midterm Exam 2025-26
4CS2-01/4IT2-01/4AID2-01/4CCS2-01/4CIT2-01
Subject-Discrete Mathematics Structure
CSE/IT/AI&DS/Cyber/IOT
17/04/2026/Friday

Roll No. 24G67CC103

Time: 2 Hours

Maximum Marks: 40

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Part A (Answer should be given up to 25 words only)

All questions are compulsory

10x1 = 10

- | | |
|---|-----------|
| Q.1 What is complemented lattice. | [CO3, L1] |
| Q.2 Define the existential and universal quantifiers. | [CO2, L1] |
| Q.3 When are two finite state automata said to be equivalent. | [CO2, L3] |
| Q.4 What is a generating function? Give an example. | [CO3, L1] |
| Q.5 How many ways are there to arrange the lattes in the word COMBINATIONS such that all vowels are together? | [CO3, L1] |
| Q.6 Determine if the set {2,4,6,8,10} from a lattice under the relation division. | [CO3, L3] |
| Q.7 Define Hamiltonian graph. | [CO5, L1] |
| Q.8 Define the Graph with example. | [CO5, L1] |
| Q.9 Define Homomorphism. | [CO4, L1] |
| Q.10 Define weighted Graph. | [CO5, L1] |

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

5 x 3 = 15

- | | |
|---|-----------|
| Q.1 Find the DNF of $\sim(p \vee q) \leftrightarrow p \wedge q$ | [CO2, L2] |
| Q.2 Consider an algebraic structure $(G, *)$ where G is the set of all non-zero real number and $*$ is binary operation defined by $a * b = \frac{ab}{4}$, show that $(G, *)$ be an abelian group. | [CO4, L4] |
| Q.3 Find the generating function for the sequence {1,1,2,3,5, 8,...} defined by Fibonacci number with $(a_0=0, a_1=1)$. | [CO3, L2] |
| Q.4 Show that for multiplication the set $G = \{1, -1, i, -i\}$ is Abel group and find its generators. | [CO4, L3] |
| Q.5 Define the following with example (a) Group (b) Graph (c) Path | [CO5, L1] |

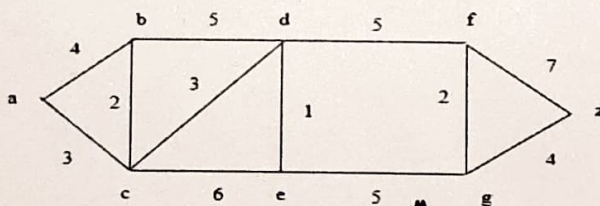
Part C (Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions (word Limit 125)

3x 5 = 15

- Q.1 Find the Shortest path between the vertex a to z.

[CO5, L2]



- Q.2. Using the generating function to solve the recurrence relation $a_r - 7a_{r-1} + 10a_{r-2} = 0$ for $r \geq 2$ with the conditions $a_0 = 10$ and $a_1 = 41$. [CO4, L3]
- Q.3. Find the CNF of $(p \wedge q) \vee (\sim p \wedge q \wedge r)$. [CO2, L2]

**GLOBAL INSTITUTE OF TECHNOLOGY****B. Tech. IV Semester I-Midterm Exam 2025-26****4CS1-02, TECHNICAL COMMUNICATION****BRANCH-CSE/ME/EE/CE****09/03/2026/ Monday****Time: 2 Hours****Maximum Marks: 40****Attempt all questions**

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. **no supplementary sheet shall be issued in any case.**

Part A (Answer should be given up to 25 words only)**All questions are compulsory**

- | | |
|---|----------|
| Q1. What is communication? | (CO1) L1 |
| Q2. Define technical writing. | (CO1) L1 |
| Q3. List the types of communication. | (CO1) L1 |
| Q4. Name the types of summaries in technical texts. | (CO2) L2 |
| Q5. Write four characteristics of effective technical writing. | (CO1) L3 |
| Q6. What do you mean by electronic communication? | (CO2) L2 |
| Q7. Define note making. | (CO2) L1 |
| Q8. What are the structural elements in reading the technical texts? | (CO2) L3 |
| Q9. Identify two barriers to effective communication. | (CO1) L2 |
| Q10. Describe the importance of written communication in organizations. | (CO1) L2 |

10 × 1 = 10**Part B Analytical/Problem solving questions****Attempt all questions (word Limit 75)**

- | | |
|---|----------|
| Q1. Explain the difference between general communication and technical communication. | (CO1) L2 |
| Q2. Explain the importance of note-making in technical communication by outline method. | (CO2) L2 |
| Q3. How can you define interpretation of the technical Text & Annotation? | (CO2) L1 |
| Q4. Discuss about technical communication skills. | (CO1) L2 |
| Q5. Explain linguistic ability in Technical Communication. | (CO1) L2 |

5 × 3 = 15**Part C (Descriptive/Analytical/Problem Solving/Design Question)****Attempt all questions (word Limit 125)**

- | | |
|---|----------|
| Q1. Explain Gardener's theory of multiple intelligence. | (CO1) L2 |
| Q2. What is comprehension of technical text? Explain its importance in technical communication. | (CO2) L2 |
| Q3. Discuss ABCS technique of technical writing. | (CO1) L2 |

3 × 5 = 15



Roll No. 24667CS103

GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. IV Semester I-Midterm Exam 2025-26
4CS4-07: Data Communication and Computer Networks
Branch-CS/AIDS/IT/Cyber Security

11/03/2026/ Wednesday

Time: 2 Hours

Maximum Marks: 40

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. **no supplementary sheet shall be issued in any case.**

MID-TERM-1

Part A (Answer should be given up to 25 words only)

All questions are compulsory

- | | |
|--|----------|
| Q.1 Define Computer Network. | (CO1) L1 |
| Q.2 List the basic components of a data communication system. | (CO1) L1 |
| Q.3 Define Telnet. | (CO1) L1 |
| Q.4 Name different network devices. | (CO1) L1 |
| Q.5 Given period is 400ms. Find the frequency in KHz. | (CO1) L2 |
| Q.6 Describe the Limitations of Data Rate. | (CO1) L2 |
| Q.7 Name any two organizations responsible for protocol standards. | (CO1) L1 |
| Q.8 Explain Periodic Analog Signal. | (CO1) L2 |
| Q.9 Calculate how the number of cables and ports are determined in Star topology and Mesh topology for a network having 8 nodes. | (CO1) L4 |
| Q.10 List the functions of the presentation Layer. | (CO1) L1 |

10x1 = 10

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

- | | |
|---|----------|
| Q.1 Define Signal Transmission. Also explain Analog and Digital Signal. | (CO1) L1 |
| Q.2 Explain the introduction of error in the Data Link Layer. | (CO2) L2 |
| Q.3 Define Logical addressing and physical addressing. | (CO2) L1 |
| Q.4 Describe transmission mode and its type. | (CO1) L2 |
| Q.5 Write various types of networks topologies in computer network. | (CO1) L3 |

5 x3 =15

Part C (Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions (word Limit 125)

- | | |
|--|----------|
| Q.1 Draw diagrams of all different Line Coding Techniques of following data: (NRZ & RZ)
101001110 | |
| Q.2 Differentiate TCP/IP and OSI with architectures. Discuss functionalities of Physical layer. | (CO1) L1 |
| Q.3 Define these terms | (CO1) L2 |
| a) Protocol & RJ45 | (CO1) L2 |
| b) Multicast & Broadcast | |
| c) Wired or Wireless Media | |

3x 5 =15



Roll No. 24C1A7C003

GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. IV Semester Mid Term II Exam 2026
4CS4-06/ Theory of Computation
Branch - CSE/IT/AI&DS /CYBER/IOT

Date/Day- 20/04/2026, Monday

Time: 2 Hours

Maximum Marks: 40

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Part A (Answer should be given up to 25 words only)

All questions are compulsory

- | | |
|---|----------|
| Q.1 Define Turing Machine with diagram | [CO4,L3] |
| Q.2 What is deciding properties of CFL? | [CO5,L2] |
| Q.3 Discuss the automata theory. | [CO5,L3] |
| Q.4 What is NOFA? | [CO4,L2] |
| Q.5 What is complexity classes? | [CO5,L1] |
| Q.6 What is pushdown automata? | [CO3,L2] |
| Q.7 What are the problems in NP complete class? | [CO5,L3] |
| Q.8 What is Acceptance by Empty Stack in PDA? | [CO3,L1] |
| Q.9 Differentiate between CNF and GNF. | [CO3,L2] |
| Q.10 What is Chomsky hierarchy? | [CO4,L3] |

10x 1 = 10

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 100)

- | | |
|--|----------|
| Q.1 Design pushdown automata (PDA) accepting $\{a^n b^m c^n \mid m, n \geq 1\}$ by null store. | [CO3,L3] |
| Q.2 Convert the following context free grammar (CFG) into an equivalent CFG in Chomsky normal form. $A \rightarrow BAB \mid B \mid \epsilon, B \rightarrow 00 \mid \epsilon$ | [CO3,L4] |
| Q.3 Construct TM for the addition function for the unary number system. | [CO4,L3] |
| Q.4. Explain Multi-tape Turing Machine and compare it with a single tape TM | [CO5,L3] |
| Q.5 Give 5 examples of languages that cannot be accepted by Finite automata but can be accepted by pushdown automata. Contrast the reason behind it. | [CO3,L2] |

5 x 3 = 15

Part C (Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions

- | | |
|---|----------|
| Q.1 Construct CFG for: <ul style="list-style-type: none">• Palindromes• Equal number of a's and b's | [CO3,L4] |
| Q.2 How the acceptance of string done in Turing machine? Construct a TM machine for checking the palindrome of the string of even length. | [CO4,L3] |
| Q.3 Discuss the problems in types of complexity classes, also compare P, NP, CO-NP, NP-hard and NP- complete. | [CO5,L4] |

3x 5 = 15

Roll No. 24667CS103**GLOBAL INSTITUTE OF TECHNOLOGY****B. Tech. IV Semester, I-Midterm Exam 2025-26****4CS4-06/4IT4-06/4AID4-06 Theory of Computation****Branch: CSE/IT/IOT/AI&DS/CYBER SECURITY****11/03/2026/ Wednesday****Time: 2 Hours****Maximum Marks: 40****Attempt all questions**

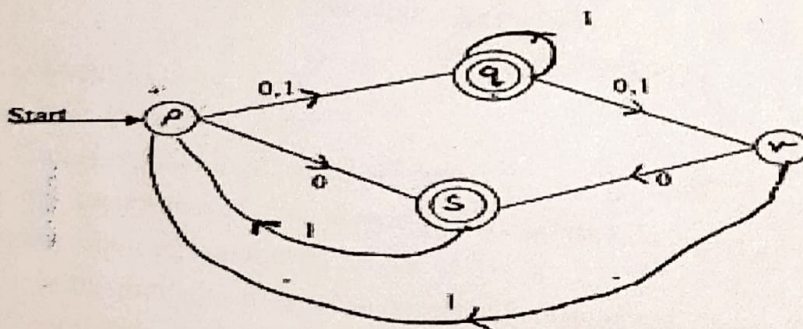
Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Part A (Answer should be given up to 25 words only)**All questions are compulsory**

- | | | |
|------|---|------------------|
| Q.1 | Find a regular expression corresponding to each of the following subset $\{0,1\}$ | [CO1, Level 1] |
| Q.2 | What is the difference between DFA and NFA? | [CO1, Level 2] |
| Q.3 | What is Kleene Closure (L^*) and Positive Closure (L^+)? Explain with example. | [CO2, Level 3] |
| Q.4 | Define Alphabet, String and Language with examples. | [CO2, Level 2] |
| Q.5 | Define the following operations on languages: <ul style="list-style-type: none">• Union• Concatenation | [CO2, Level 2] |
| Q.6 | Explain the finite automata with Epsilon Transition. | [CO1, Level 2] |
| Q.7 | Describe the method of subset construction to convert a given NFA into equivalent DFA with suitable example. | [CO1, Level 1] |
| Q.8 | Explain about the closure properties of regular languages. | [CO2, Level 1] |
| Q.9 | What are the regular operators applied to the regular languages? Explain it. | [CO1, Level 2] |
| Q.10 | Draw the finite automata
$(10+01)^*0(001+011)^*$ | [CO2, Level 4] |
| | | 10×1 = 10 |

Part B Analytical/Problem solving questions**Attempt all questions (word Limit 75)**

- Q.1 Convert the following NFA into equivalent DFA.



- Q.2 Convert the following NFA into equivalent DFA using subset construction and also show the transition diagram for this DFA. [CO2, Level 5]



Roll No. 24261JCC103

GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. IV Semester II midterm Exam 2025-26
4CS3-04/ Microprocessor and Interfaces
Branch: CSE/AI&DS/CC/CY

18/03/2025, Saturday

Time: 2 Hours

Maximum Marks: 40

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. **no supplementary sheet shall be issued in any case.**

Part A (Answer should be given up to 25 words only)

All questions are compulsory

- | | |
|--|------------|
| Q.1 Describe the Stack And which instruction are used for Stack? | (CO2, L-2) |
| Q.2 Define Instruction cycle. | (CO2, L-1) |
| Q.3 State the difference between jump and call instruction. | (CO2, L-1) |
| Q.4 What are the different mode of 8255PPI. | (CO4, L-1) |
| Q.5 What are the difference between delay and counter. | (CO3, L-1) |
| Q.6 Describe the use of RIM & SIM instruction. | (CO3, L-2) |
| Q.7 Explain LHLD & LDA instruction. | (CO2, L-2) |
| Q.8 Difference between vectored and non-vectored interrupt. | (CO3, L-4) |
| Q.9 List the technique which are used to provide Time delay. | (CO3, L-1) |
| Q.10 Define the RET instruction. | (CO2, L-1) |

10x 1 = 10

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

- | | |
|--|------------|
| Q.1 Write a Program to multiply Two 8-bit numbers? | (CO3, L-3) |
| Q.2 What is interrupt? Explain types of interrupts. | (CO3, L-2) |
| Q.3 What is Subroutine? How it is use in 8085 microprocessors. | (CO3, L-2) |
| Q.4 Write a Program to transfer a block of data in Direct order. | (CO3, L-3) |
| Q.5 Explain the Pin diagram of 8259. | (CO3, L-2) |

5 x 3 =15

Part C (Descriptive/Analytical/Problem Solving/Design Question)

Attempt all questions (word Limit 125)

- | | |
|---|------------|
| Q.1 Draw and explain the architecture of 8259(PIC). | (CO3, L-3) |
| Q.2 Draw and explain the Timing Diagram of MVI B,08H. | (CO2, L-3) |
| Q. Draw and explain the architecture of 8255. | (CO4, L-3) |

3x 5 =15



Roll No. 24 EGTCS103

GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. IV Semester I midterm Exam 2025-26
4CS3-04/ Microprocessor and Interfaces
Branch: CSE/AI&DS/CC/CY

10/03/2025, Tuesday

Time: 2 Hours

Maximum Marks: 40

Attempt all questions

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. **no supplementary sheet shall be issued in any case.**

Part A (Answer should be given up to 25 words only)

All questions are compulsory

- | | |
|--|-------------|
| Q.1 What are the basic units of a microprocessor? | (CO1, L-1) |
| Q.2 Difference Between Static & Dynamic RAM. | (CO1, L-2)) |
| Q.3 Define Program Counter. | (CO1, L-1) |
| Q.4 What is Stack & what is the function of stack Pointer. | (CO1, L-1) |
| Q.5 Let the content of Accumulator is 21H and content of Register B is 46H, If instruction
ADD B is executed then specify the result. | (CO-2, L-2) |
| Q.6 Mention interrupts in 8085 microprocessors. | (CO1, L-1) |
| Q.7 Define instruction cycle, machine cycle, T-state. | (CO2, L-1) |
| Q.8 Explain the difference between microprocessor and microcontroller. | (CO-1, L-2) |
| Q.9 Explain MOV & MVI instruction. | (CO2, L-2) |
| Q.10 What is the function of ALE signal in 8085 microprocessors. | (CO1, L-1) |

10x 1 = 10

Part B Analytical/Problem solving questions

Attempt all questions (word Limit 75)

- | | |
|--|------------|
| Q.1 Explain Different types of Addressing mode of 8085 microprocessor? | (CO2, L-2) |
| Q.2 What are the different flags in 8085 microprocessors? Explain it. | (CO1, L-2) |
| Q.3 Draw the Bus organization of 8085 microprocessor. Explain all types of Buses. | (CO2, L-3) |
| Q.4. Explain the need to demultiplex the bus AD ₀ -AD ₇ . | (CO1, L-2) |
| Q.5 Classify 8085 microprocessor instructions on the basis of their length (size). | (CO1, L-2) |

5 x 3 =15



GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. IV Semester I Midterm Exam 2025-26
4CS4-05-Database Management System
Branch: CSE, AI&DS, IT, IOT, CYBER SECURITY
10/03/2026- Tuesday

Time: 2 Hours**Maximum Marks: 40****Attempt all questions**

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Part A (Answer should be given up to 25 words only)**All questions are compulsory**

- | | | |
|------|--|-----------|
| Q.1 | List the main differences among Primary Key, Candidate Key, and Foreign Key. | [CO1, L1] |
| Q.2 | Explain why database systems are preferred over file systems. | [CO1, L2] |
| Q.3 | Define the terms Entity and Entity Set in an ER model. | [CO1, L1] |
| Q.4 | Compare DDL and DML with suitable examples. | [CO1, L4] |
| Q.5 | Distinguish between Strong entity set and Weak entity set. | [CO1, L4] |
| Q.6 | Illustrate Composite and Multivalued attributes with examples. | [CO1, L3] |
| Q.7 | Define Data Independence | [CO1, L1] |
| Q.8 | Define Cardinality Mapping in a relation. | [CO1, L1] |
| Q.9 | List the types of participation constraints. | [CO1, L1] |
| Q.10 | Justify whether a Primary Key can contain NULL values. | [CO1, L5] |
- 10×1 = 10**

Part B Analytical/Problem solving questions**Attempt all questions (word Limit 75)**

- | | | |
|-----|--|-----------|
| Q.1 | Define Unary, Binary, and Ternary relationships in an ER model. | [CO1, L1] |
| Q.2 | Construct two sample relations and demonstrate the results of Union, and Intersection, operations | [CO2, L6] |
| Q.3 | Analyze how the three-schema architecture helps in achieving data abstraction. | [CO1, L4] |
| Q.4 | Evaluate which architecture is more suitable for large web applications and justify your answer. | [CO1, L5] |
| Q.5 | Illustrate Generalization and Specialization by drawing an ER diagram for a Vehicle system (Car, Bike, Truck). | [CO1, L3] |

5×3 = 15**Part C (Descriptive/Analytical/Problem Solving/Design Question)****Attempt all questions (word Limit 125)**

- | | | |
|-----|---|-----------|
| Q.1 | Design and draw a complete ER diagram for an Online Shopping System, clearly indicating entities, relationships, primary keys, key constraints, and participation constraints. | [CO1, L6] |
| Q.2 | Student (roll_no INT, name VARCHAR (50), age INT, department_id INT)
Department (department_id INT, department_name VARCHAR(50))
Course (course_id INT, course_name VARCHAR (50), department_id INT)
Write SQL queries for the following:
a) Retrieve the names of students who are in the "Computer Science" department.
b) List all courses offered by the "Electronics" department.
c) Create all three tables with proper constraints.
d) Change the age of student with roll_no = 105 to 22 | [CO2, L3] |
| Q.3 | Illustrate the types of relationships in an ER model by constructing ER examples for One-to-One, One-to-Many, and Many-to-Many relationships. | [CO1, L3] |

Roll No. 24C6TCS103

GLOBAL INSTITUTE OF TECHNOLOGY
B. Tech. IV Semester II Midterm Exam 2025-26
4CS4-05/ Database Management System
Branch: CSE, AI&DS, IT, IOT, CYBER SECURITY
18.04.2026/Saturday

Time: 2 Hours**Maximum Marks: 40****Attempt all questions**

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. No supplementary sheet shall be issued in any case.

Part A (Answer should be given up to 25 words only)**All questions are compulsory**

- | | | |
|------|---|---------|
| Q.1 | Define trivial and non-trivial functional dependencies with suitable examples | CO3, L2 |
| Q.2 | What is the purpose of schema refinement | CO3, L2 |
| Q.3 | Define closure of an attribute set. | CO3, L2 |
| Q.4 | Describe the ACID properties. | CO4, L1 |
| Q.5 | Explain serializability in transaction processing. | CO4, L2 |
| Q.6 | Define deadlock in DBMS with a suitable example. | CO5, L2 |
| Q.7 | Describe the phases of validation based protocol. | CO5, L1 |
| Q.8 | Define timestamp in DBMS and state its purpose | CO5, L2 |
| Q.9 | Describe dirty read problem. | CO5, L2 |
| Q.10 | How a database failure can be avoided? | CO5, L2 |

10x 1 = 10**Part B Analytical/Problem solving questions****Attempt all questions (word Limit 100)**

- | | | |
|-----|---|---------|
| Q.1 | Differentiate conflict and view serializability with an example. | CO4, L4 |
| Q.2 | Explain transaction states with diagram. | CO4, L2 |
| Q.3 | Describe briefly with Suitable examples.
A) Deadlock Handling
B) Recoverable Schedules
C) Phantom phenomenon
D) Cascadeless schedules | CO4, L2 |
| Q.4 | Explain shadow paging recovery protocol along with its pros and cons | CO5, L2 |
| Q.5 | Differentiate between lock-based and timestamp-based concurrency control techniques. | CO5, L4 |

5 x 3 =15**Part C (Descriptive/Analytical/Problem Solving/Design Question)****Attempt all questions**

- | | | |
|-----|---|---------|
| Q.1 | what is log? Discuss the silent features of deferred database modification and immediate database modification in brief | CO5, L4 |
| Q.2 | Discuss Strict Two-Phase Locking (Strict 2PL). How is it different from basic 2PL? | CO4, L5 |
| Q.3 | Check whether the given schedule S is conflict Serializable or not. If yes, then determine all the possible serialized schedules- | CO5, L4 |

T1	T2	T3	T4
	R(A)		R(A)
W(B)		R(A)	
	W(A)		
	W(B)	R(B)	

3x 5 =15